

Austin Moon

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EDUCATION

University of Waterloo

B.A.Sc. Mechatronics Engineering, Specialization in Artificial Intelligence

Waterloo, ON

Sep. 2024 – Expected June 2029

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, VHDL

AI/ML & CV: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, NumPy, pandas, Matplotlib

Embedded/Tools: STM32, Arduino, I2C/SPI/UART, CAN bus, Git, Linux, VS Code

Certificates: Machine Learning Specialization – Supervised/Unsupervised/Reinforcement Learning

EXPERIENCE

Embedded Software Developer

May 2025 – Aug 2026

Waterloo Aerial Robotics Group

Waterloo, ON

- Developed a robust C++ driver for a 3-axis magnetometer on STM32, using I2C communication with DMA to read magnetic flux density and compute a real-time compass heading
- Extracted and parsed position, altitude, and speed from the u-blox M10 GPS receiver using an interrupt-driven UART NMEA stream and converted it into navigation coordinates
- Implemented a low-pass filter in C for the ICM-42688 IMU within WARG's firmware, reducing noise in sensor data
- Configured an MCP3004 ADC with full-duplex SPI to analyze 10-bit analog signals for precise data analysis

Mechatronics Engineering Intern

Sep. 2025 – Dec. 2025

Spyder Controls Corporation

Lacombe, AB

- Implemented an Arduino Nano Python script to validate thermistors by comparing analog voltage readings and flagging out-of-tolerance ($\pm 5\%$) resistance values, open circuits, and short circuits
- Flashed and validated firmware and circuits on 10+ products, systems, and test jigs over a CAN bus network
- Led a UV chamber rebuild project, building and wiring the control box, LED chamber, and cooling system, with integration of multiple power supplies, relays, DAC circuitry, and sensor inputs for safe, reliable operation

Mechanical Engineer

Jan. 2025 – Apr. 2025

University of Waterloo

Waterloo, ON

- Supervised the Rapid Prototyping Centre and the Engineering Student Machine Shop, assisting 40+ students per day on 3D printing and laser cutting as well as AutoCAD and SolidWorks design
- Set up and operated the Rayjet R500 laser cutter and Prusa/Bambu 3D printers, resolving technical issues

PROJECTS

NFL Defensive Coverage Predictor | *Python, PyTorch, ByteTrack, YOLOv11, OpenCV* | [GitHub](#)

- Trained a custom YOLOv11 object detection model on a football helmet dataset with 10,000+ images, achieving 69% mAP50 and 75% precision, with ByteTrack for multi-object tracking across occlusions
- Engineered a 339-dimensional feature vector from raw NFL tracking data (4.8M rows, 18 weeks, 11.5K plays) and trained a PyTorch MLP to classify 6 defensive coverage types using player positioning and formation indicators
- Built an end-to-end video-to-prediction inference pipeline with automated field calibration using Hough line transforms for yard-line and hash-mark detection, coordinate mapping, and snap-frame detection

NBA Game Predictor | *Python, NumPy, Pandas, Scikit-learn* | [GitHub](#)

- Built a Python ETL pipeline that uses a REST API to pull NBA team and player statistics for the 2025–26 season, then cleans, validates, and exports datasets to CSV using pandas for downstream modeling
- Engineered rolling-average features and compared Logistic Regression, Random Forest, and Gradient Boosting classifiers on 1000+ games, achieving 63% accuracy (vs. 54% majority-class baseline)
- Implemented a Tkinter dashboard for selecting dates and generating game-specific predictions

G-Helper Clone | *Python, PyQt6, D-Bus* | [GitHub](#)

- Developed a Python + PyQt6 system tray application for managing ASUS laptop hardware on Fedora Linux, leveraging Claude AI as a development assistant throughout the build
- Tuned 50+ kernel/hardware parameters (CPU EPP, PCIe ASPM, PCI runtime PM) reducing idle power by 80%
- Built a debounced state machine to auto-configure CPU frequency, GPU DPM, and fan PWM on battery power
- Engineered a real-time hwmon sensor system with Tdie/Tctl register readings and battery discharge computation